Postman POC

Proof of Concept for Postman API Testing

By **Mahmood Ahmad**

Scope



Evaluate **Postman** for API testing.



Highlight **features, cost, integrations, pros & cons**.



Assess feasibility for **team adoption**.

Key Features



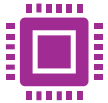
Teams & workspaces



Environments & Variables



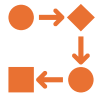
Bulk Execution



CI/CD



Flows (No-Code API architecture)



API Chaining



Performance Testing



Security & Secure Vault



PostBot AI



Reporting



Unit Testing



Monitoring



Mocking



Integrations

Teams & Workspaces



Personal & Team Workspaces – Organize API collections (**All Plans**).

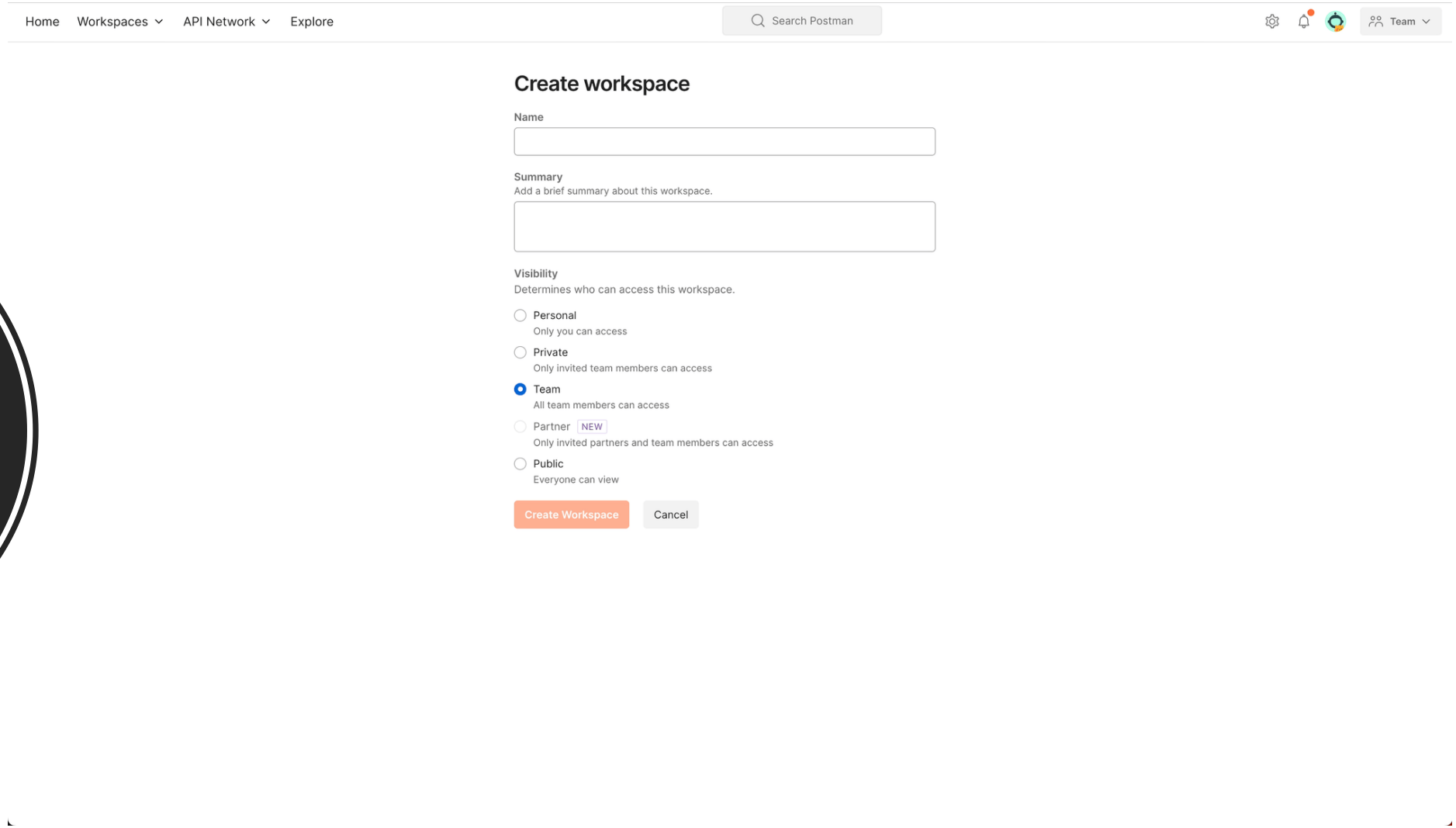


Public & Private Workspaces – Secure or share API testing (**Professional, Enterprise**).



Internal & Multi-Partner Workspaces – Controlled team collaboration (**Professional, Enterprise**).

Workspaces


Home
Workspaces
API Network
Explore
Search Postman
Team
Create workspace
Name
Summary
Add a brief summary about this workspace.
Visibility
Determines who can access this workspace.
Personal
Only you can access
Private
Only invited team members can access
Team
All team members can access
Partner
NEW
Only invited partners and team members can access
Public
Everyone can view
Create Workspace
Cancel

Environments & Variables



Global, Environment, Collection Variables – Reusable data storage (**All Plans**).

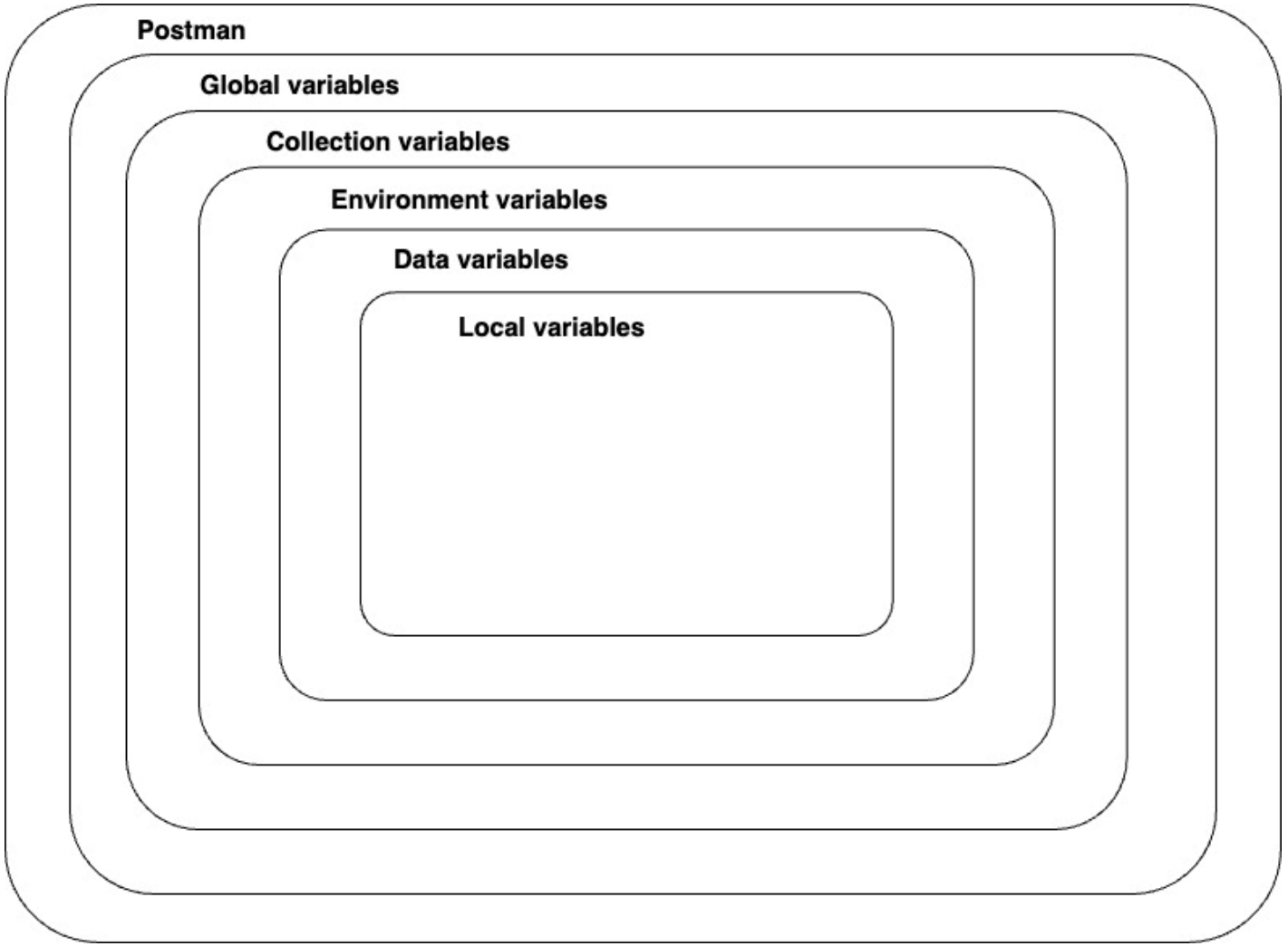


Path Variables, Query Params, Headers – Parameterized API calls (**All Plans**).



Secret Variables & Vault – Store credentials securely (**Enterprise**).

Environment Variables



API Chaining



Pass Data Between API Requests – Extract and reuse response values (**All Plans**).



Use Pre-request & Test Scripts – Store dynamic data in variables (**All Plans**

end-to-end



Unit Testing (Chai.js)

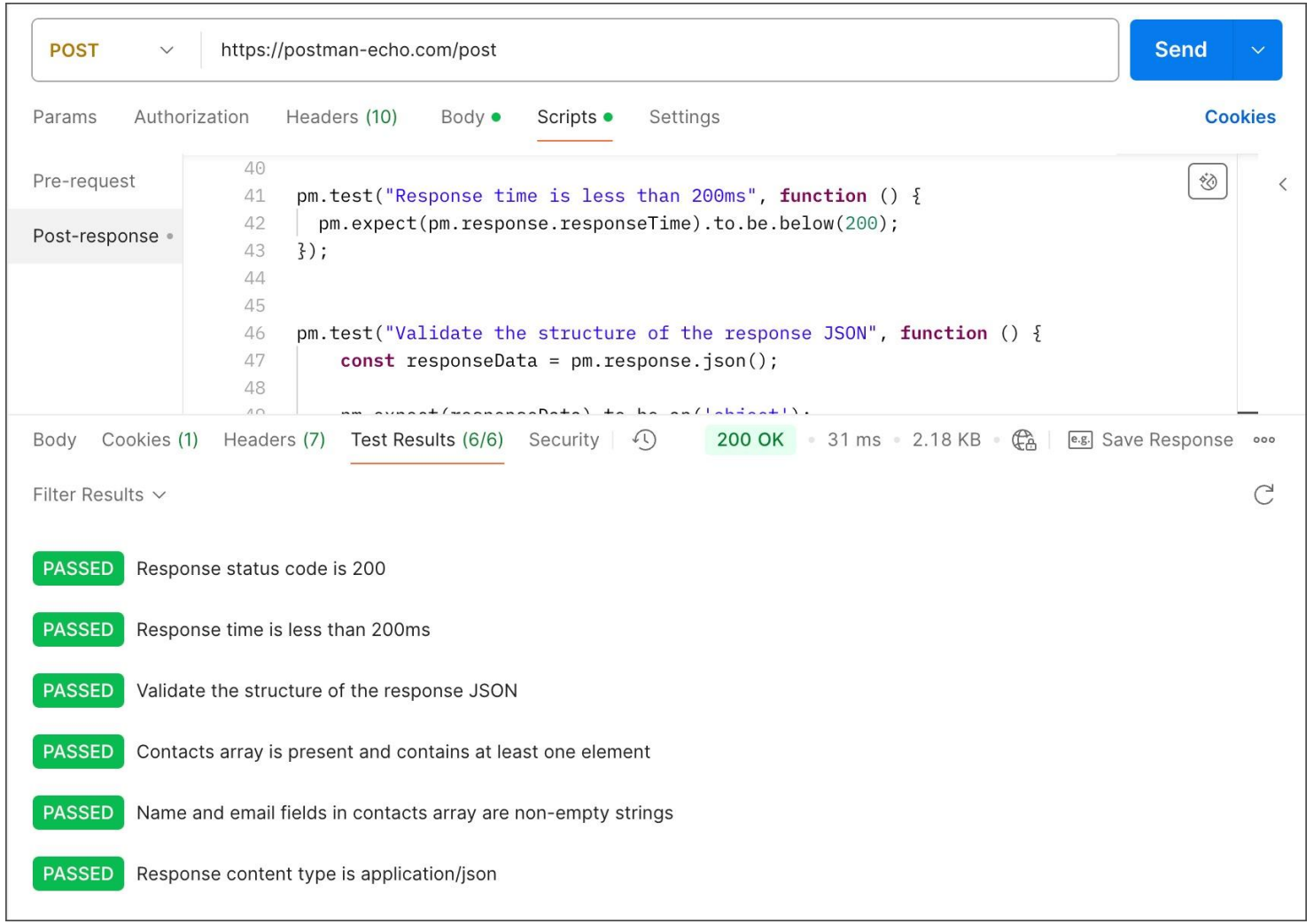
Assertion-Based Testing – Validate API responses (**All Plans**).

Supports BDD & TDD Approaches (**All Plans**).

Pre-Built Snippets for Chai.js (**All Plans**).

Generate Test Data – use faker.js library to generate random date of different types **(All Plans)**

Use scripts to add logic and tests



Performance Testing

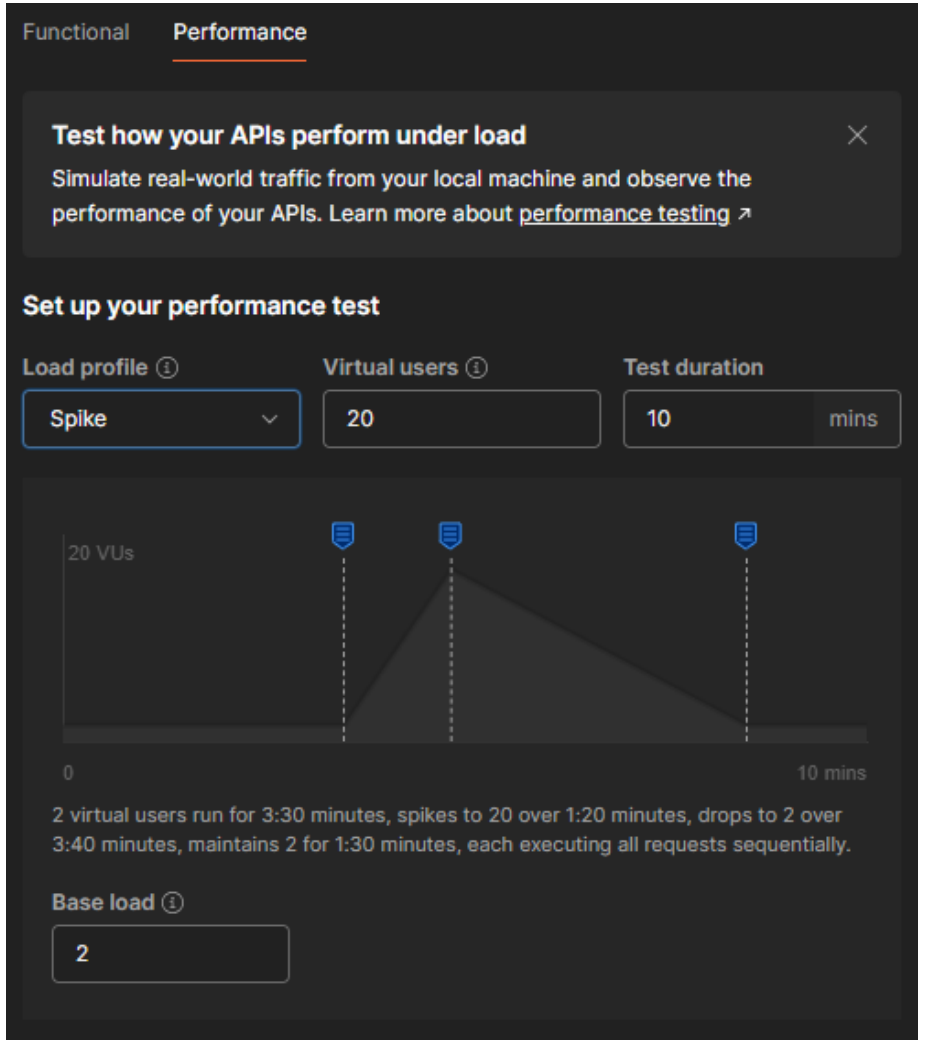


Simulate High API Loads – Identify bottlenecks (**Professional, Enterprise**).



Run Virtual User Tests – Measure response time & scalability (**Professional, Enterprise**).

performance



Bulk Execution



Run Collections Manually – Execute API tests interactively (**All Plans**).

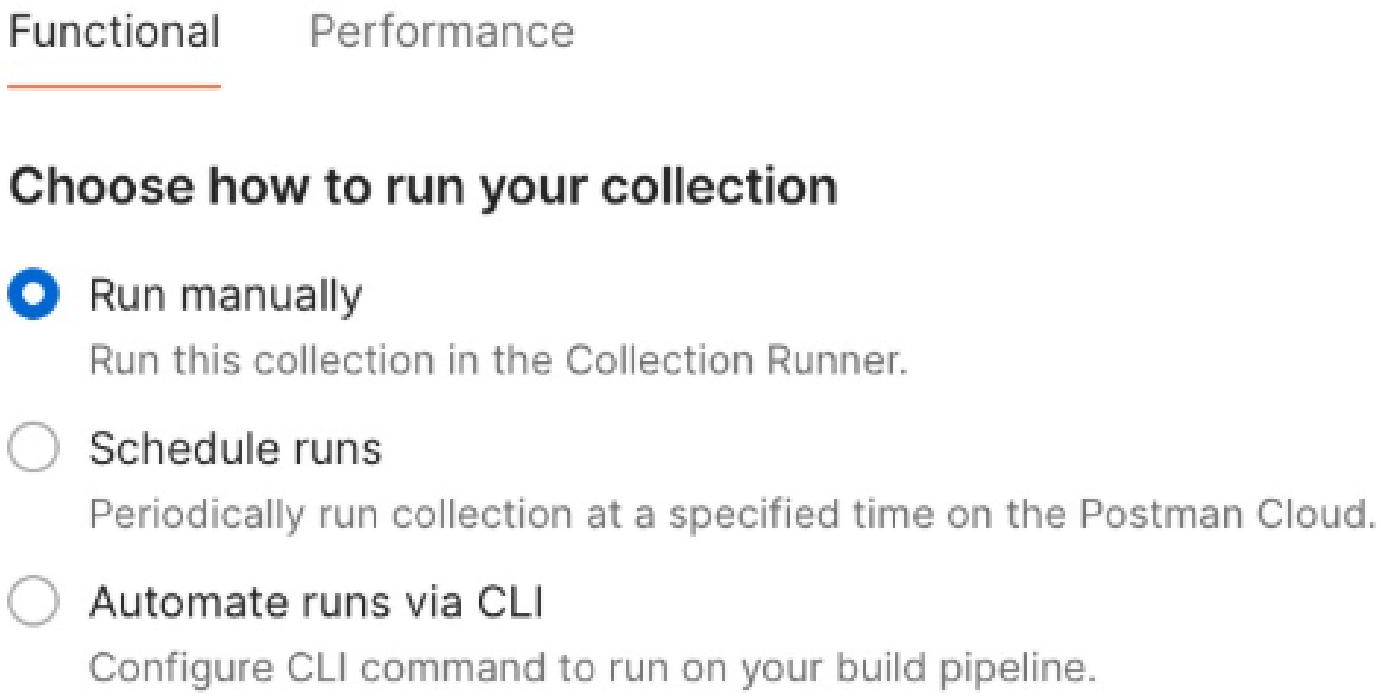


Scheduled Execution – Automate test runs (**Basic, Professional, Enterprise**).



Postman CLI & Newman Support – Run tests via command line (**Basic, Professional, Enterprise**).

Collection Runner



Reporting



Real-time test execution results – with assertion-level details.

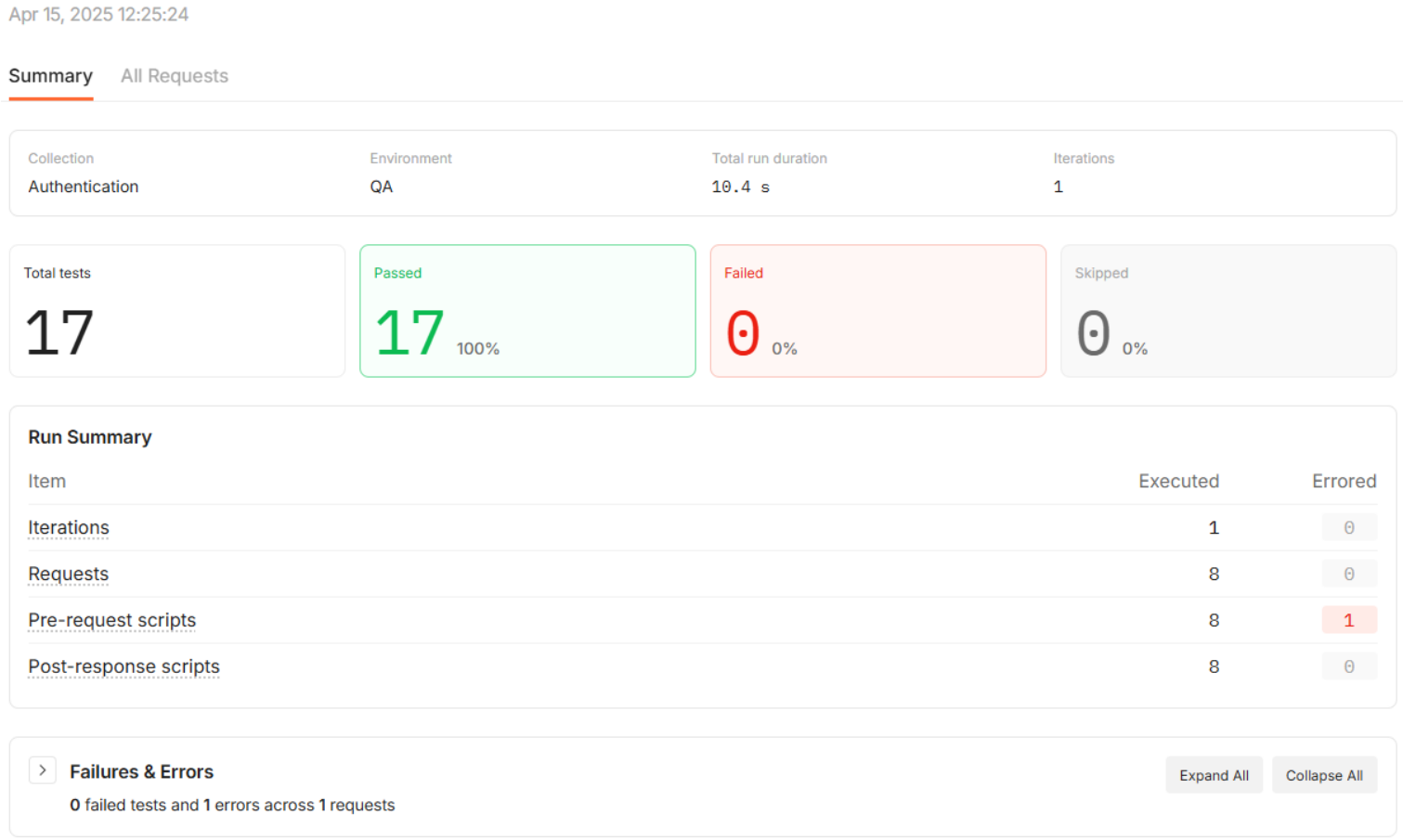


Newman CLI – enables multi-format reporting (CLI, JSON, HTML, JUnit) for automation use.



Test logs – include the requests sent, response codes and times, and the number of passed and failed tests

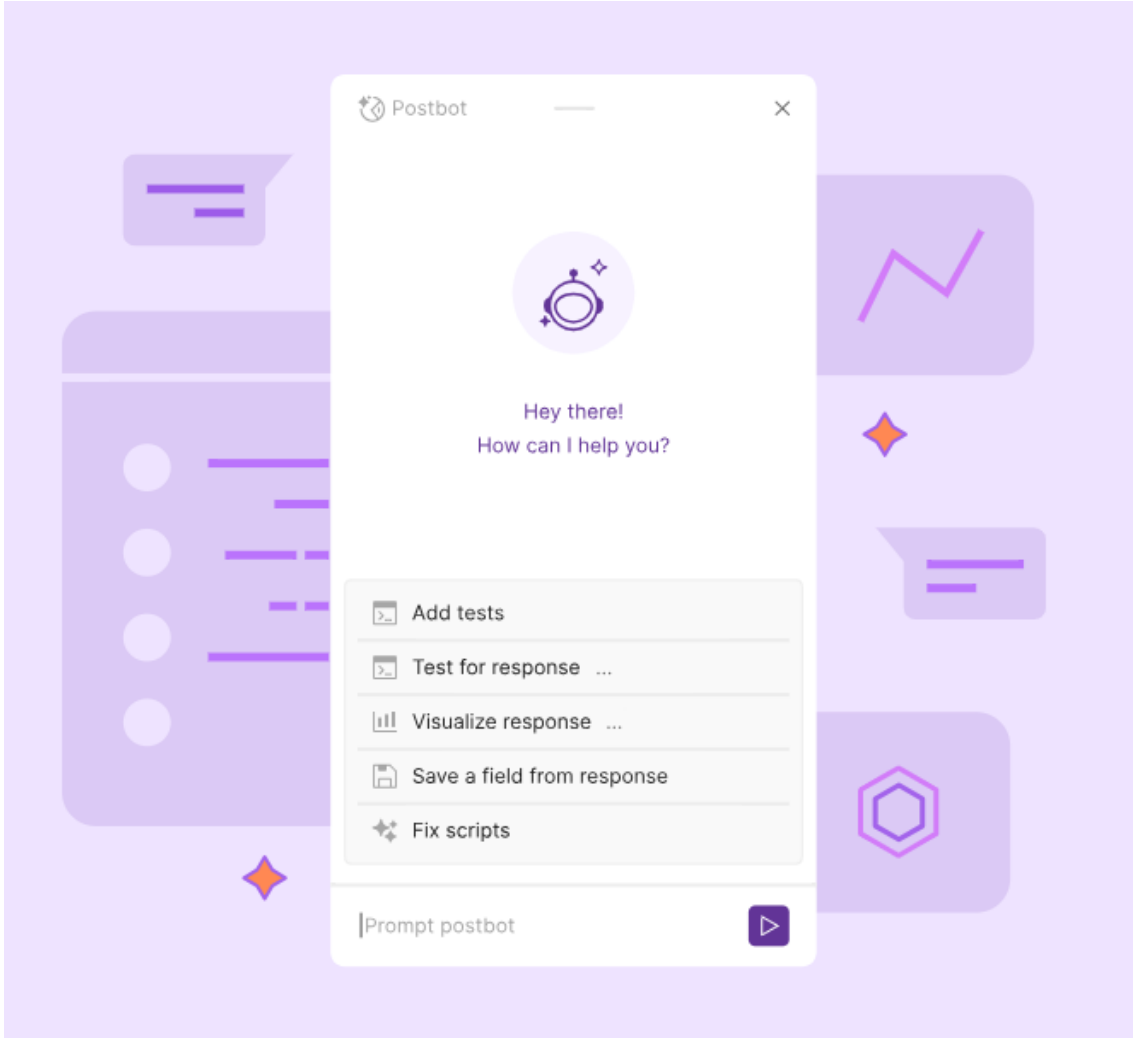
Generate collection run reports using the Postman CLI | Postman Docs



Postbot AI

- **Test Case Generation** – Auto-generates test scripts using Chai.js.
- **Auto-Suggested Assertions** – Detects response patterns and suggests validation checks.
- **Natural Language Commands** – Converts plain English queries into test scripts.
- **API Documentation & Insights** – Extracts and structures API details.
- **Visualize Response** – Analyze API response and create visual representations of date

Postbot


Postbot
Hey there!
How can I help you?
Add tests
Test for response ...
Visualize response ...
Save a field from response
Fix scripts
Prompt postbot

Monitoring



Check API health and performance - Ensure the API is up and running in production and other environments.



Run tests in multiple environments and regions - Ensure your APIs are working everywhere. (Running monitors in multiple regions requires a paid Postman plan.)

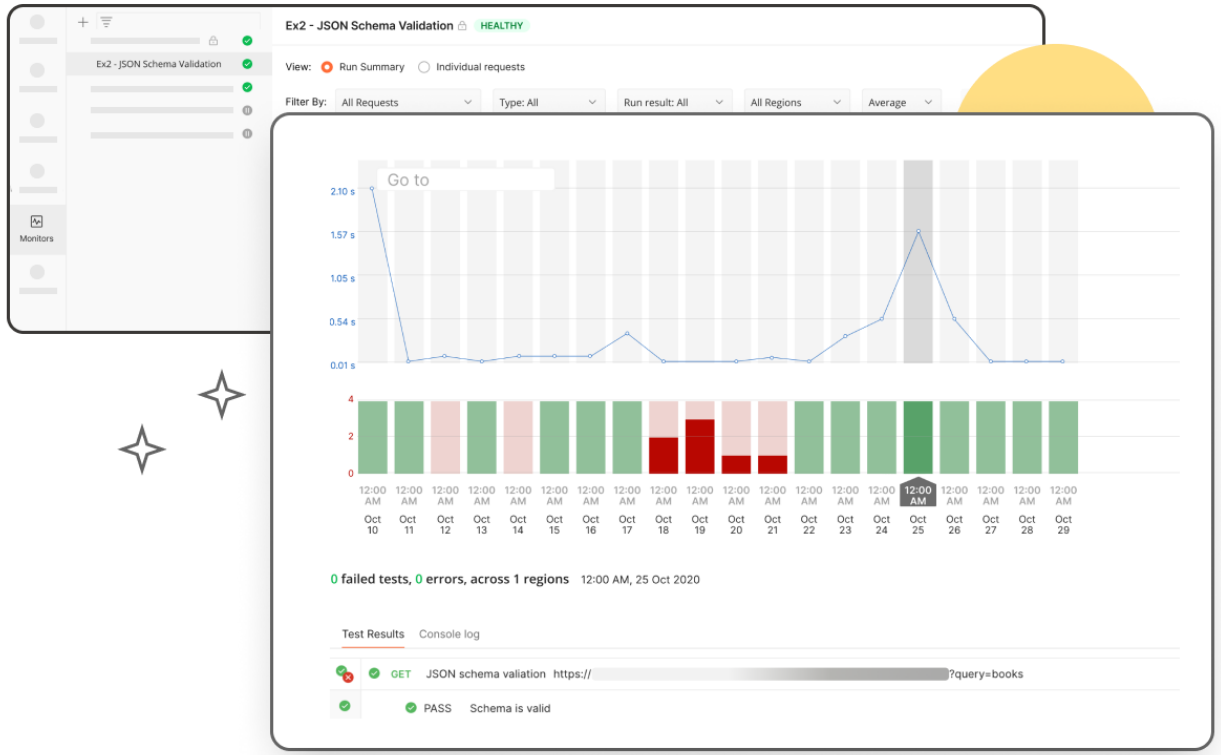


Visualize results on the monitor dashboard - Get better visibility into API performance over time and identify trends.



Monitor the security of your endpoints - Continuously test APIs for known security vulnerabilities.

Monitor


Ex2 - JSON Schema Validation
HEALTHY
Ex2 - JSON Schema Validation
Monitors
View: Run Summary Individual requests
Filter By: All Requests Type: All Run result: All All Regions Average
Go to
2.10 s
1.57 s
1.05 s
0.54 s
0.01 s
4
2
0
12:00 AM Oct 10
12:00 AM Oct 11
12:00 AM Oct 12
12:00 AM Oct 13
12:00 AM Oct 14
12:00 AM Oct 15
12:00 AM Oct 16
12:00 AM Oct 17
12:00 AM Oct 18
12:00 AM Oct 19
12:00 AM Oct 20
12:00 AM Oct 21
12:00 AM Oct 22
12:00 AM Oct 23
12:00 AM Oct 24
12:00 AM Oct 25
12:00 AM Oct 26
12:00 AM Oct 27
12:00 AM Oct 28
12:00 AM Oct 29
0 failed tests, 0 errors, across 1 regions 12:00 AM, 25 Oct 2020
Test Results
Console log
GET JSON schema valiation https:// ?query=books
PASS Schema is valid

Security & Vault



External Vault Support – 1Password, AWS Secrets Manager, Azure Key Vault, HashiCorp Vault (**Enterprise**).

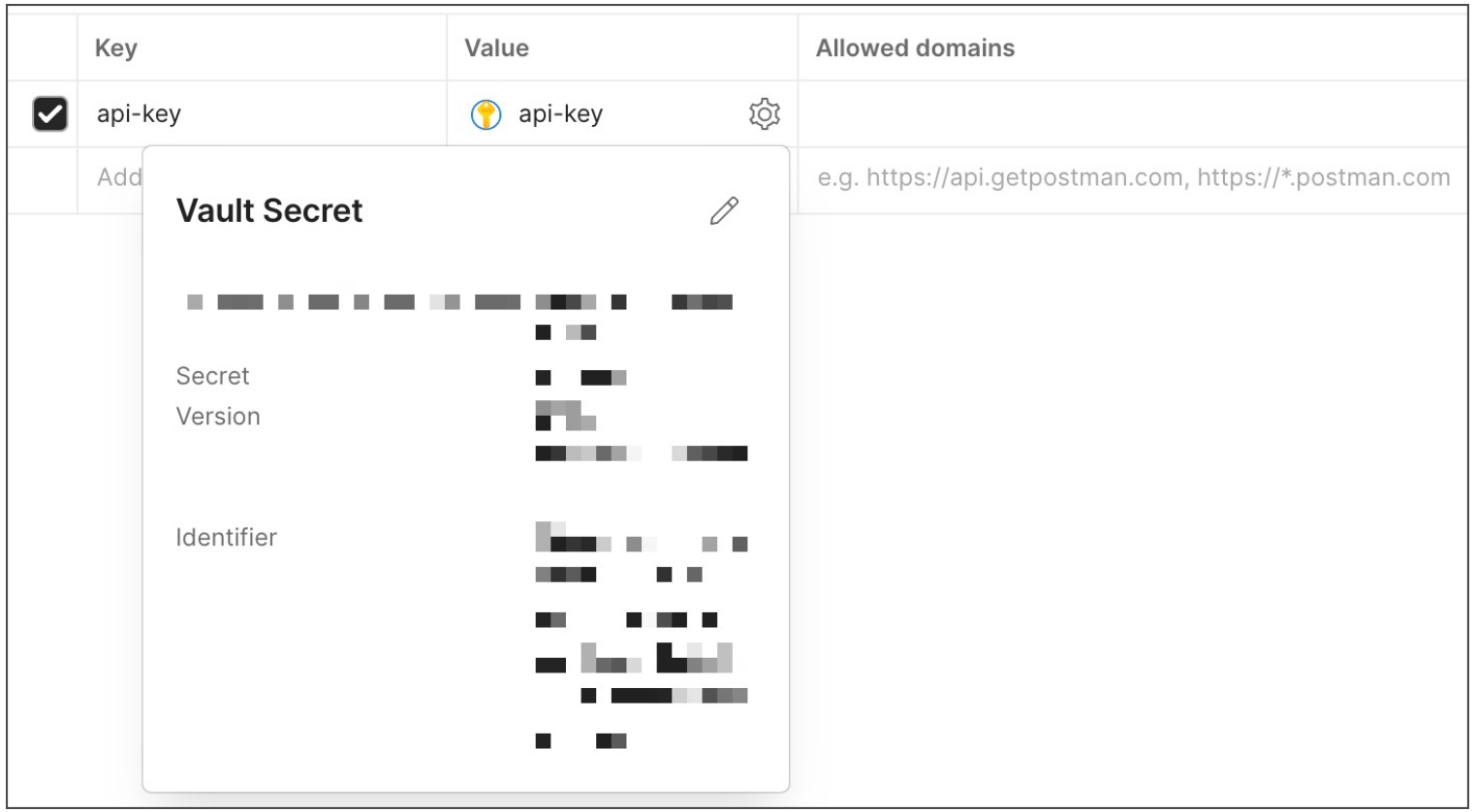


Postman Proxy & Interceptor – Capture & inspect API requests (**Professional, Enterprise**).



SSO, SCIM, & SAML – Advanced authentication (**Enterprise**).

Postman Vault

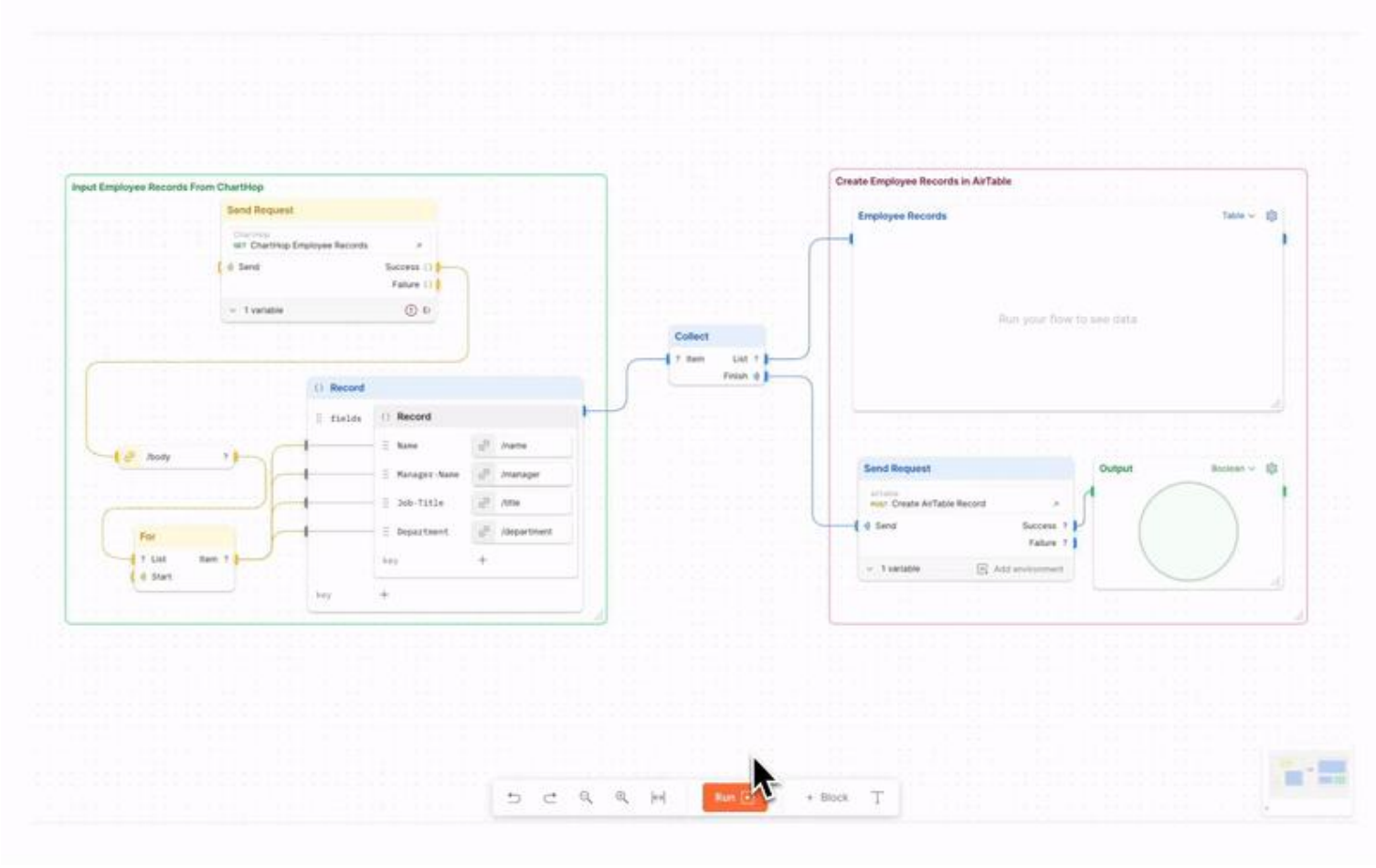


Flows (No-Code API architecture)

Drag & Drop Workflow Builder – Automate API requests (**All Plans**).

Advanced Automation with Logic – Conditional flows and loops (**Professional, Enterprise**).

Flows


Input Employee Records From ChartHop
Send Request
ChartHop Employee Records
Send
Success
Failure
1 variable
Record
fields
Record
Name
Manager Name
Job Title
Department
/name
/manager
/title
/department
/body
For
List
Item
Start
Collect
Item
List
Finish
Create Employee Records in AirTable
Employee Records
Table
Run your flow to see data
Send Request
Create AirTable Record
Send
Success
Failure
1 variable
Add environment
Output
Boolean
Run
Block

Mock servers



simulate API responses – based on your saved examples, allowing frontend or testing teams to work without a live backend.



Unique mock URL – that can be called from anywhere, enabling testing with predefined request-response pairs.

Mock Servers



CI/CD

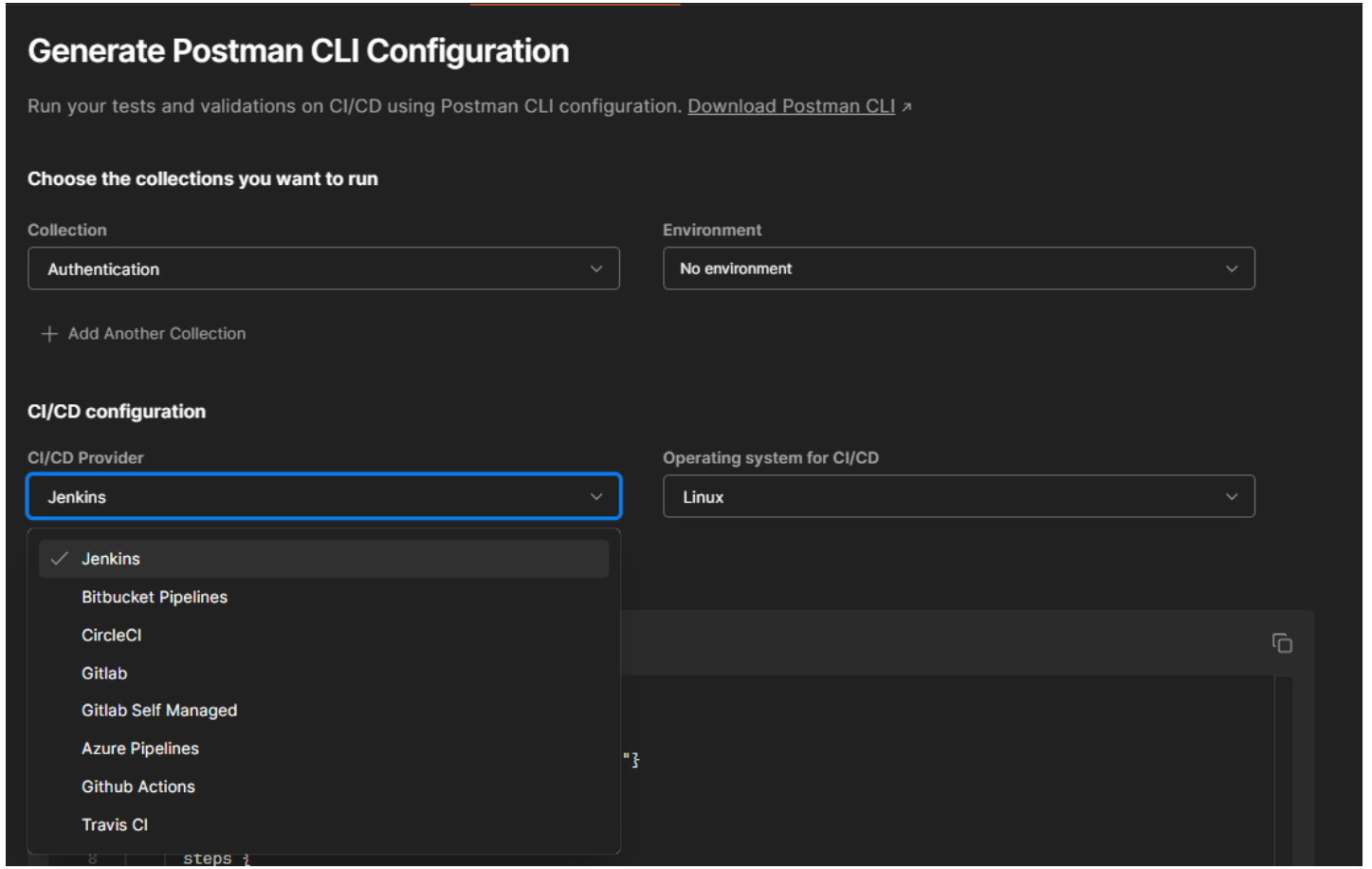


GitHub, GitLab, Bitbucket – Version control (**Basic, Professional, Enterprise**).



Jenkins, Azure DevOps, GitHub Actions – Automated API testing in pipelines (**Basic, Professional, Enterprise**).

Integrate CI



Integrations

- Automate tests via Jenkins, GitHub Actions, GitLab, etc.
- Sync collections with GitHub or Bitbucket.
- Enables API monitoring, documentation, and visualization.
- Connects with Slack, Microsoft Teams, and VS Code for streamlined teamwork.

Integrations



Pricing Breakdown (Feature-Specific Limits)

Feature	Free	Basic	Professional	Enterprise
Collaborators	Up to 3	Unlimited	Unlimited	Unlimited
Packages	3	3	25	100
Collection Runs	25	25	25	25
Performance Testing (Virtual Users)	100	100	500	500
Performance Testing (Test Runs)	25	25	250	250
Mock Server Requests	1,000	10,000	10,000	100,000
Monitoring Requests	1,000	10,000	10,000	10,000
Collection Recovery	1 Day	30 Days	90 Days	90 Days
Private API Network	❌	❌	❌	✅
Advanced Role-Based Access	❌	❌	✅	✅
User Groups & Customer Success	❌	❌	❌	✅
Super Admin Role	❌	❌	❌	✅
Collection Viewers	5	10	25	1:1

Pricing Breakdown (Additional Add-ons)

Feature	Pricing
Postbot	$9 per seat/month
Unlimited Collection runs	$49 per seat/month
Security & Vault Integrations	$29 per seat/month
Cloud Integrations (Slack, Teams)	$14 Basic Plan
Additional Mocks	$0.75 / 1000 calls
Flows	$12 per seat/month

Pros of using Postman for API Testing

- **User-Friendly Interface** – No need for scripting; intuitive UI for sending API requests, debugging, and test automation.
- **Comprehensive API Testing** – Supports environments, variables, request chaining, mock servers, and automated testing within a single platform.
- **Collaboration & Version Control** – Teams can work together in shared workspaces with role-based access, Git integration, and real-time collaboration.
- **CI/CD & DevOps Integration** – Seamlessly integrates with Jenkins, GitHub, Azure DevOps, and AWS for test automation in pipelines.
- **Built-in API Documentation & Mocking** – Generates interactive API documentation and supports mock server creation without external tools.
- **Security & Authentication Support** – Supports OAuth, API keys, encrypted secrets, and integration with AWS, Azure, and HashiCorp vaults.
- **Cloud Test data Storage** – Provides cloud storage to store test data files and access remotely during executions
- **Response Visualization & Reporting** – Offers detailed response views, including raw data, formatted JSON, and response time analysis.

Cons of using Postman for API Testing

- **Limited Load & Performance Testing** – Cannot generate large-scale concurrent virtual users like JMeter; maxes out at **500 virtual users**.
- **GUI differs CLI:** Results & configurations may vary between GUI and CLI runs.
- **No database testing** – Postman does not have direct **JDBC** support for validating backend database queries.
- **Low code reusability**: Packages can be used to reuse customer utility methods. But reusing requests is not an option.
- **Limited AI Efficiency** – AI generated tests through Postbot may need **manual adjustments** for **complex scenarios**.
- **Pricing Tiers** – Essential features are mostly **paid**, with **limited** free **access** to key functionalities.
- **Limited Security Testing** – No built-in **vulnerability** scanning, **fuzz** testing, or **encryption** analysis.
- **Incomplete Documentation** – Not everything is covered in the official documentation.